

Silvia Coffee Machine - User Manual



Welcome to Your Silvia Coffee Machine

Your Silvia Coffee Machine combines traditional espresso craftsmanship with modern touchscreen technology. This manual will guide you through operating your machine safely and effectively to create exceptional coffee.



Quick Start Guide

First Time Setup

1. **Power On:** Ensure machine is plugged in and powered on
2. **Wait for Connection:** Look for “CONNECTED” status in bottom right
3. **Initial Heating:** Machine automatically starts heating to brew temperature
4. **Ready to Brew:** When temperature reaches target, you’re ready to make coffee

Making Your First Espresso

1. **Prepare:** Fill water reservoir, insert portafilter with ground coffee
2. **Start Brewing:** Press the **BREW** button on home screen
3. **Wait for Temperature:** Watch the temperature gauge reach target
4. **Place Cup:** Position cup on scales, wait for scales to settle
5. **Begin Extraction:** Press the ► **BREW NOW** button
6. **Monitor Progress:** Watch the real-time extraction charts

7. **Stop When Done:** Press **II STOP** when desired volume reached



Home Screen Overview

Main Interface Elements

Temperature Gauge (Right Side)

- **Large Circular Display:** Shows current boiler temperature
- **Color Coding:**
 - Blue: Normal temperature
 - Red: Over target temperature
- **Interactive:** Tap to access temperature settings

Operation Buttons (Left Side)

BREW Button

- Starts heating to brew temperature
- Transitions to brew screen
- Only active when connected

STEAM Button

- Starts heating to steam temperature
- Used for milk steaming
- Shows **II stop** button when active

FLUSH Button

- Runs cleaning cycle
- Flushes water through system
- Shows **II** stop button when active

Status Indicators (Bottom)

Connection Status

- Green “CONNECTED”: Hardware communication active
- Red “DISCONNECTED”: Check hardware connection

Emergency Stop

- Red button always accessible
- Immediately stops all operations
- Use in emergency situations only



Brewing Process

Step-by-Step Brewing

1. Preparation Phase

- Fill water reservoir with fresh, filtered water
- Grind coffee beans to appropriate fineness
- Dose and tamp coffee in portafilter
- Insert portafilter into group head

2. Temperature Control

- Machine automatically heats to brew temperature (default 93°C)
- Monitor temperature on circular gauge
- Adjust target temperature in settings if needed

3. Pre-Brewing Setup

- Press **BREW** button to enter brew mode
- Place cup on scales platform
- Wait for scales to settle (stable weight reading)
- Ensure “scales settled” indicator is active

4. Extraction Process

- Press ► **BREW NOW** when temperature is reached
- Scales automatically tare (zero) at start
- Monitor real-time extraction data:
- **Coffee Weight:** Grams of extracted coffee
- **Pressure:** Extraction pressure in bar
- **Time:** Extraction duration
- Watch live charts showing extraction curves

5. Stopping Extraction

- Press || **STOP** when desired volume reached
- Typical espresso: 25-30g in 25-30 seconds
- Machine automatically stops pump and closes valve

Understanding the Brew Screen

Top Information Bar

- **Back Button:** Return to home screen (stops brewing)
- **Boiler Temp:** Current temperature with target comparison
- **Pump Power:** Current pump power percentage
- **Extraction Time:** Live brewing timer

Real-Time Charts

Coffee Weight Chart (Top)

- Shows cumulative coffee weight over time
- Green line indicates extraction rate
- Auto-scaling based on extraction volume

Pressure Chart (Bottom)

- Shows extraction pressure over time
- Red line indicates pressure profile
- Typical range: 8-10 bar for espresso

Control Buttons

- **► BREW NOW:** Start extraction (only when ready)
- **|| STOP:** Stop current operation

Brewing Tips

Optimal Extraction Parameters

- **Dose:** 18-20g ground coffee
- **Yield:** 25-35g extracted coffee
- **Time:** 25-35 seconds
- **Pressure:** 8-10 bar
- **Temperature:** 90-96°C

Reading the Charts

- **Steep Rise:** Good initial extraction
- **Gradual Curve:** Consistent flow rate
- **Flat Line:** Extraction complete or channeling
- **Pressure Spikes:** Possible grind too fine



Steam Mode

Using Steam Function

Starting Steam Mode

1. Press **STEAM** button on home screen
2. Machine heats to steam temperature (default 130°C)
3. Higher temperature required for steam production
4. Wait for temperature to reach target

Steaming Milk

1. Pour cold milk into steaming pitcher

2. Insert steam wand into milk
3. When temperature reached, steam is available
4. Use traditional steaming techniques
5. Press **II STOP** when finished

Steam Safety

- Steam temperature is much higher than brew temperature
- Always use caution around steam wand
- Clean steam wand after each use
- Never leave steam mode unattended

Flush Mode

Cleaning Cycles

When to Flush

- Before first use of the day
- After extended periods of non-use
- When switching between different coffees
- As part of regular maintenance

Running Flush Cycle

1. Remove portafilter from group head
2. Place container under group head

3. Press **FLUSH** button
4. Machine runs pump at full power
5. Fresh water flushes through system
6. Press **II STOP** when complete

Flush Benefits

- Removes coffee residue
- Heats up brewing components
- Ensures fresh water in system
- Maintains optimal taste

Settings and Customization

Accessing Settings

- Tap the temperature gauge on home screen
- Settings screen opens with current values
- Adjust parameters as needed
- Press **SAVE** to confirm changes

Temperature Settings

Brew Temperature

- **Range:** 60°C - 110°C
- **Default:** 93°C

- **Light Roasts:** 94-96°C
- **Dark Roasts:** 90-92°C
- **Adjustment:** $\pm 1^\circ\text{C}$ increments

Steam Temperature

- **Range:** 110°C - 150°C
- **Default:** 130°C
- **Higher:** More steam pressure
- **Lower:** Gentler steaming

Scale Management

Tare Function

- **Purpose:** Zero the scales
- **When:** Before each brewing session
- **How:** Press **TARE** button in settings
- **Automatic:** Scales auto-tare when brewing starts

Calibration

- **Purpose:** Ensure accurate weight readings
- **When:** If weights seem inaccurate
- **How:** Press **CAL** button, follow prompts
- **Weight:** Use known calibration weight (100-500g)

Saving Settings

- All temperature changes automatically saved
- Settings persist between sessions
- Calibration data stored permanently
- Press **BACK** to return without saving



Understanding Data and Charts

Real-Time Monitoring

Temperature Display

- **Current:** Live boiler temperature
- **Target:** Desired temperature setting
- **Color:** Red when over target
- **Accuracy:** $\pm 0.1^{\circ}\text{C}$ precision

Pressure Readings

- **Range:** 0-16 bar
- **Brewing:** Typically 8-10 bar
- **Pre-infusion:** Lower pressure start
- **Over-extraction:** Pressure too high

Weight Measurements

- **Precision:** $\pm 0.1\text{g}$ accuracy
- **Dual Sensors:** Two load cells for stability

- **Auto-tare:** Automatic zeroing
- **Settling:** Waits for stable reading

Chart Interpretation

Extraction Curves

- **Ideal Shape:** Smooth, consistent curve
- **Fast Start:** Good coffee preparation
- **Steady Flow:** Proper grind size
- **Sudden Stop:** Possible channeling

Pressure Profiles

- **Gradual Rise:** Good pre-infusion
- **Stable Plateau:** Consistent extraction
- **Pressure Drops:** Flow rate changes
- **Spikes:** Grind or dose issues



Safety and Troubleshooting

Safety Features

Automatic Safety Systems

- **Overheat Protection:** Stops heating at maximum temperature
- **Timeout Protection:** Stops operations after maximum time
- **Communication Monitoring:** Stops if connection lost

- **Emergency Stop:** Manual override always available

Safety Indicators

- **Temperature Warnings:** High temperature alerts
- **Error Messages:** Clear problem descriptions
- **Connection Status:** Hardware communication status
- **Operation Timeouts:** Maximum time limits

Common Issues and Solutions

“DISCONNECTED” Status

Problem: No communication with hardware

Solutions:

- Check power connections
- Verify USB cable connection
- Restart application
- Contact support if persistent

Temperature Not Rising

Problem: Boiler not heating

Solutions:

- Check power supply
- Verify heater connections
- Check for error messages

- Allow more time for heating

Inaccurate Weight Readings

Problem: Scales showing wrong values

Solutions:

- Tare the scales
- Remove items from platform
- Calibrate with known weight
- Check for mechanical interference

No Water Flow

Problem: Pump running but no water

Solutions:

- Check water reservoir level
- Verify pump connections
- Check for blockages
- Prime the pump system

Pressure Too High/Low

Problem: Extraction pressure incorrect

Solutions:

- Adjust grind size (finer = higher pressure)
- Check dose amount

- Verify tamp pressure
- Clean group head

Error Messages

Hardware Errors

- **PT100_FAULT**: Temperature sensor issue
- **ADS1115_INIT_FAILED**: Pressure sensor problem
- **COMM_TIMEOUT**: Communication lost

Operation Errors

- **OVERHEAT**: Temperature too high
- **BREW_TIMEOUT**: Brewing time exceeded
- **STEAM_TIMEOUT**: Steam time exceeded

User Errors

- **NOT_CONNECTED**: Hardware not available
- **INVALID_STATE**: Operation not allowed in current state
- **SCALES_NOT_SETTLED**: Wait for stable weight reading

When to Contact Support

- Persistent connection issues
- Hardware error messages
- Unusual temperature readings

- Mechanical problems
- Safety concerns

Maintenance and Care

Daily Maintenance

After Each Use

- Wipe down exterior surfaces
- Clean steam wand if used
- Empty drip tray if full
- Check water reservoir level

End of Day

- Run flush cycle
- Turn off machine
- Clean portafilter and basket
- Wipe down all surfaces

Weekly Maintenance

Water System

- Empty and refill water reservoir
- Clean reservoir with mild soap
- Check for mineral buildup

- Run extended flush cycle

Cleaning

- Deep clean portafilter and baskets
- Clean group head screen
- Wipe down all surfaces
- Check for loose connections

Monthly Maintenance

Descaling

- Use appropriate descaling solution
- Follow manufacturer's instructions
- Run multiple flush cycles after descaling
- Check all seals and gaskets

Calibration Check

- Verify scale accuracy with known weights
- Check temperature accuracy if possible
- Recalibrate if necessary
- Update any drift in readings

Professional Service

- Annual professional inspection recommended

- Pressure testing of boiler system
- Electrical safety inspection
- Replacement of wear items

Support and Resources

Getting Help

Built-in Help

- Error messages provide specific guidance
- Status indicators show system state
- Charts help diagnose extraction issues

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- Technical manual for detailed information
- Installation guide for setup issues
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Community Support

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Contact Information

- **Technical Support:** [Contact details]

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- **Emergency Support:** [Contact details]

Warranty Information

- Standard warranty covers manufacturing defects
- Proper maintenance required for warranty coverage
- Professional installation recommended
- Keep purchase receipt for warranty claims



Enjoy Your Coffee!

Your Silvia Coffee Machine is designed to help you create exceptional espresso with precision and consistency. Take time to experiment with different settings and techniques to find your perfect cup.

Remember: Great coffee is a combination of quality beans, proper preparation, and precise extraction. Your machine provides the precision - the rest is up to your creativity and skill!

Happy Brewing! ☕

⚠ **Safety Reminder:** Always follow safety guidelines when operating your machine. Never leave the machine unattended during operation. If you notice any unusual behavior, use the emergency stop and contact support. # Silvia Coffee Machine - User Manual



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Brewing Process

Step-by-Step Brewing

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Reading the Charts

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Cleaning Cycles

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Pressure Profiles

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Safety and Troubleshooting

Safety Features

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Problem: Scales showing wrong values

Solutions:

- Tare the scales
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- Calibrate with known weight
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Solutions:

- Check water reservoir level
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Error Messages

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When to Contact Support

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- Hardware error messages
- Unusual temperature readings
- Mechanical problems
- Safety concerns

Maintenance and Care

Daily Maintenance

After Each Use

- Wipe down exterior surfaces
- Clean steam wand if used
- Empty drip tray if full
- Check water reservoir level

End of Day

- Run flush cycle
- Turn off machine
- Clean portafilter and basket
- Wipe down all surfaces

Weekly Maintenance

Water System

- Empty and refill water reservoir
- Clean reservoir with mild soap
- Check for mineral buildup
- Run extended flush cycle

Cleaning

- Deep clean portafilter and baskets
- Clean group head screen
- Wipe down all surfaces

- Check for loose connections

Monthly Maintenance

Descaling

- Use appropriate descaling solution
- Follow manufacturer's instructions
- Run multiple flush cycles after descaling
- Check all seals and gaskets

Calibration Check

- Verify scale accuracy with known weights
- Check temperature accuracy if possible
- Recalibrate if necessary
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Professional Service

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Support and Resources

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